

ARNAV MALHOTRA

Student Researcher & Engineer

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RESEARCH & PATENTS

System and Method for Dynamic State Transposition and Preemptive Evasion of Spatiotemporal Correlated Errors

Patent Pending

- This filing presents a novel method for mitigating the impact of spatiotemporal correlated errors (nonrandom quantum noise coming from sources like cosmic rays and radioactivity) in quantum processors. Standard post hoc correction, while working well with random errors, suffers great difficulty in correcting correlated errors. This method involves detection of correlated errors before their impact is materialized and the transportation of the quantum information outside of the predicted path of error propagation, offering a step closer to fault-tolerant quantum computing.

High Fidelity Synthetic Financial Universes Through Regime Switching With Stochastic Volatility and Jump Diffusion (fsynth)

Preprint

- This paper introduces fsynth, a multi-factor synthetic engine that integrates a Heston-based stochastic volatility model with Merton jump diffusion, governed by a hidden Markov regime-switching process. By using macro-regime states alongside company specific "genes," the engine generates plausible universes containing both price action and fundamental accounting data.

Empirical Tests for a Hedging Framework for the Quantum Binomial Options Pricing Model

Preprint

- This paper derives and tests analytical formulae for hedging with the Quantum Binomial Options Pricing Model. The simulation reveals that the Quantum Binomial Options Pricing model is, at least in its current state, not viable for usage in actually hedging derivatives. This is due to numerical instabilities, particularly in the Quantum Theta and Quantum Vega. We conclude that the single period Quantum Binomial Options Pricing Model is currently unsuitable for practical applications of hedging.
- Presented at Terra NYC STEM Fair, March 1st, 2026

PROJECTS & OPEN SOURCE

yfinance (Library Contributor) | 30M+ Monthly Downloads

- Contributor to yfinance, the industry-standard Python library for accessing Yahoo! Finance market data.

EVE Online Market Strategies

- Used for making ISK in the game EVE online, a popular MMORPG known for its intricate economy closely resembling the real world. There is an intraregional arbitrage calculator, as well as a market price prediction and opportunities calculator.

Wharton Global High School Investment Competition

- Lead quantitative researcher and developer on team for the Wharton Investment Competition. Current finalists (top 10 of 6300 teams).

EXPERIENCE

The Stuyvesant Spectator

New York, NY

Web Department Lead

- I maintain the digital infrastructure for an award-winning high school publication. I lead the web department in optimizing the publishing of articles, preserving site reliability, and continuing 24/7 uptime.

Nurse AI Project

New York, NY

Software Member

- Working under the mentorship of Dr. Priyank Jain to create a generative AI tool to aid nurses in rural parts of West Bengal, India.

EDUCATION

Stuyvesant High School

New York, NY

Class of 2028 (Sophomore)

- Courses:** AP Calculus BC, AP Precalculus, AP Statistics, AP Human Geography, AP Environmental Science